Remote Sensing and Satellite Imagery Project

AI-powered Cloud Masking

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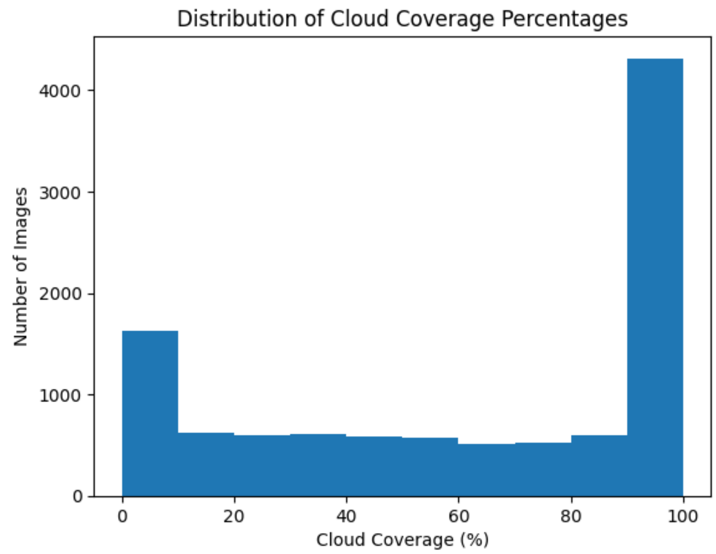
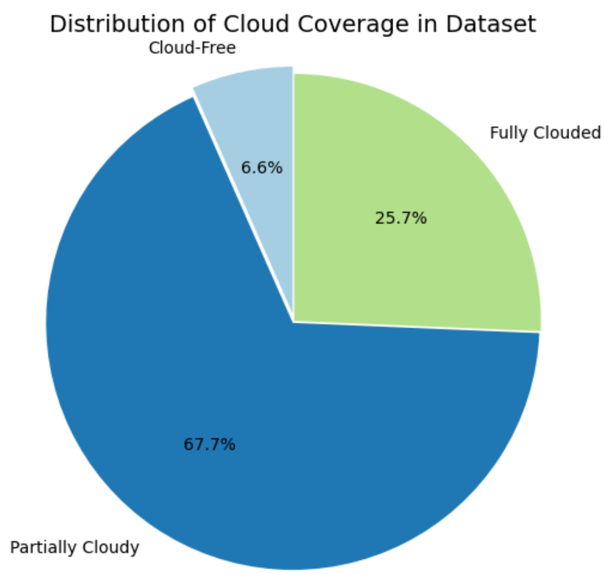
Exploratory Data Analysis

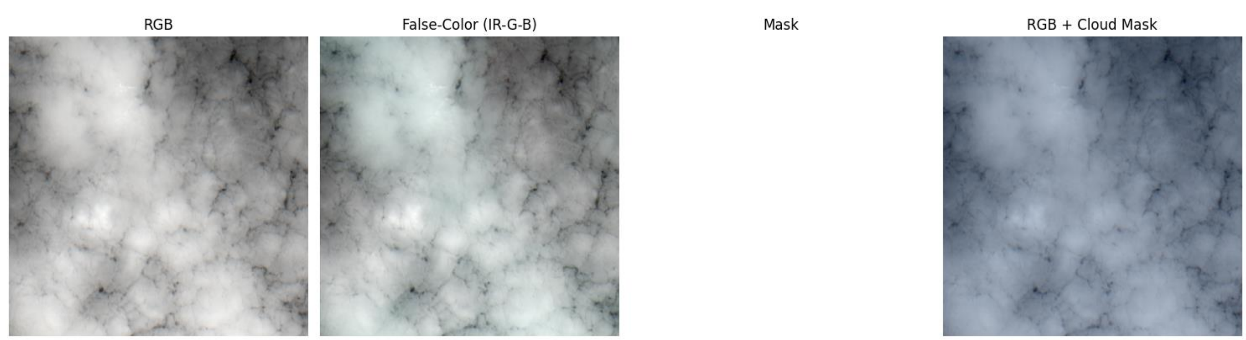


The dataset contains 10573 images of size 512x512 where each image has 4 spectral bands (channels) which are Red, Green, Blue, and IR.

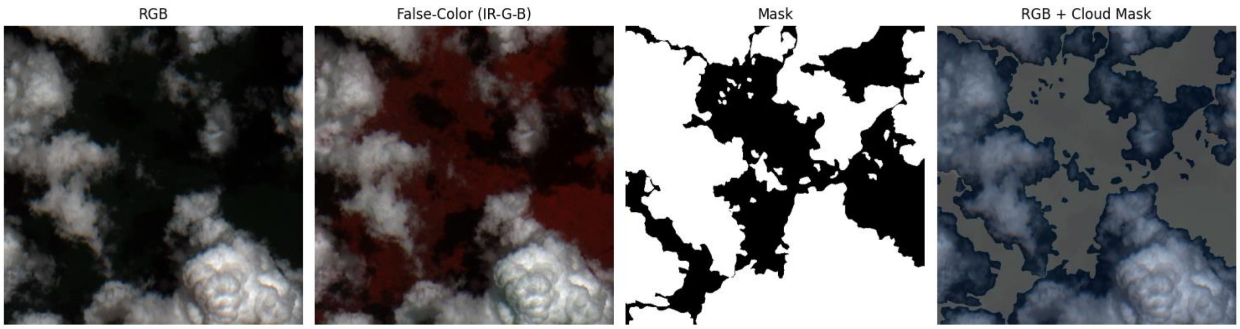
We computed the cloud coverage percentage for each image and visualized the distribution. The data is heavily skewed with a strong bias toward fully overcast images (100% coverage).



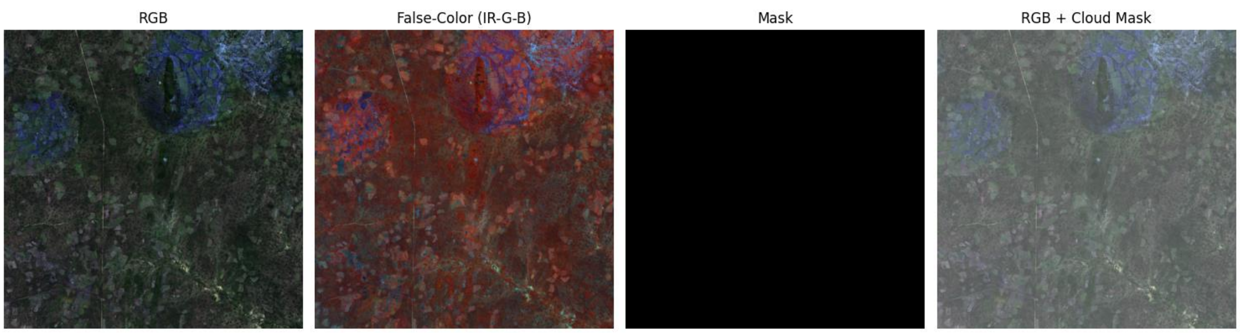
However, uponinspecting images labeled as 100% cloud coverage, we found instances of missegmented masks, for example:



The segmentation masks were mostly releable for patially clouded, for example:



and clear-sky cases, for example:



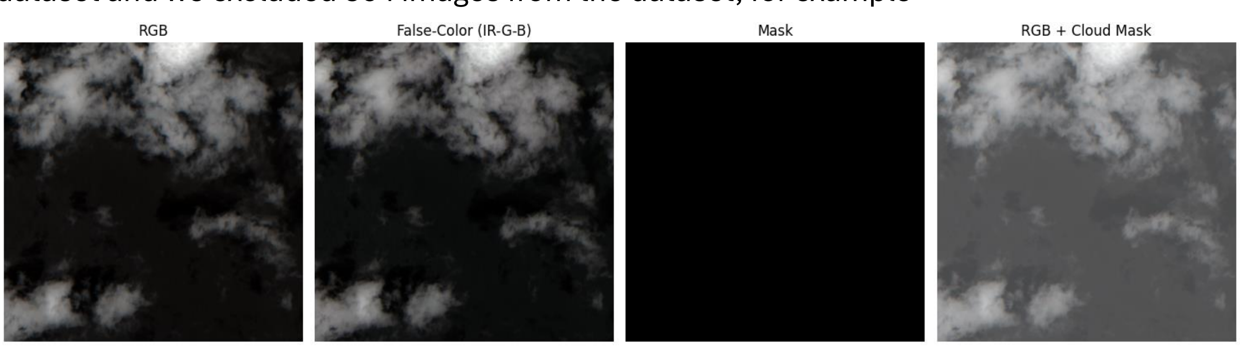
We computed per-channel statistics across the entire dataset to enable proper normalization during preprocessing. The results are:

	Red	Green	Blue	IR
Mean	2855.27820701	2845.4572077	2747.33356059	3662.23889358
Std deviation	3161.46447944	2903.18784174	2793.4662976	2427.58756888

Data Preprocessing

Manual Discard

We did a manual selection of images with missegmented masks to be excluded from the dataset and we excluded 864 images from the dataset, for example



Normalization

To standardize the input data and improve model convergence, we applied z-score normalization to each channel independently. The normalization was computed as:

$$value = \frac{\text{pixel value} - \mu_{channel}}{\sigma_{channel}}$$

where $\mu_{channel}$ and $\sigma_{channel}$ represent the mean and standard deviation of each spectral band, respectively.

Data Augmentation Experiments

After experimentation, the following augmentation pipeline yielded the best model performance by enhancing generalization while preserving critical spatial features:

- **Geometric Transformations (Training Only)**
    - **Random Flips**: Horizontal and vertical flips (50% probability each).
    - **90° Rotations**: Applied randomly (50% probability).
    - **Arbitrary Rotations**: Limited to ±30° (30% probability) with nearest-neighbor interpolation to prevent mask distortion.
    - **Random Crops**: 448×448 patches resized back to 512×512 (30% probability), encouraging scale invariance.
- **Validation Protocol**
    - No augmentations were applied to validation data beyond tensor conversion, ensuring fair evaluation on unmodified samples.

The selected transformations diversified the training set without introducing destructive noise

Band Selection

The model was trained using all four available spectral channels to maximize the utilization of spatial and spectral information.

Data Split

We split the data to train, validation, and test datasets with the following sizes

Train	Validation	Test
7189 images	1798 images	1586 images

Models Trials

Classical Models

- Implementation Challenges
    - Traditional models faced significant limitations with full-resolution multispectral data
    - Memory Constraints → Unable to process full 512×512×4 images (16MB per sample)
    - Computational Limits → Training on entire dataset exceeded available RAM
- Logistic Regression (SGD-optimized)
    - Trained on the whole train set with batch size = 1000
    - With resizing the images, dice score = 0.84
    - Without resizing, dice score = 0.85
- Random Forest
    - Trained only on a sample with size 1000 from the data
    - We made a grid search to get the best parameters
    - Obtained dice score = 0.88

Deep Learning Models

1. **CloudNet (Encoder-Decoder Architecture)**

We began with straightforward U-Net style architecture as our baseline model. The architecture was adapted from the work of Mo et al. (GitHub repository)(paper), with key modifications to accommodate our multi-spectral input data and computational constraints.

- U-Net Inspired Design:
    - Symmetric encoder-decoder structure with skip connections
    - Four-stage down sampling/up sampling pipeline (512px → 32px bottleneck)
    - Double convolutional blocks with BatchNorm and ReLU activations
- Original Contributions:
    - Input channel adaptation (3-channel RGB → 4-channel multispectral)
    - Adjusted kernel dimensions for our 512×512 input resolution
    - Modified output head for binary segmentation (cloud vs. background)

- Model Summary:

```
Total params: 1,944,193
Trainable params: 1,944,193
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 4.00
Forward/backward pass size (MB): 809.00
Params size (MB): 7.42
Estimated Total Size (MB): 820.42
```

2.**CloudNet Improved (Hybrid attention-residual architecture)**

Building on the baseline, we introduced:

- Residual Blocks:
  - Replaced plain convolutions with residual connections
  - Added dropout (p=0.3) for regularization
- Attention Gates:
  - Implemented between decoder and skip connections
  - Dynamic feature refinement using gated attention
- Advanced Components:
  - Bilinear upsampling instead of transposed convolutions
  - Dilated convolutions in bottleneck (dilation=2)
  - Increased channel dimensions (32→512)
- Model Summary:

```
Total params: 8,511,493
Trainable params: 8,511,493
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 4.00
Forward/backward pass size (MB): 3185.31
Params size (MB): 32.47
Estimated Total Size (MB): 3221.78
```

3. **CloudNet Deep Supervision**

To further enhance learning, we implemented:

- Deep Supervision:
    - Auxiliary outputs at multiple decoder levels
    - Multi-scale upsampled predictions (64px, 128px, 256px)
- Training Optimization:
    - Combined loss from all supervision heads
    - Progressive feature refinement through the network
- Architectural Refinements:
    - Balanced channel growth (16→256)
    - Dedicated upsampling layers for each supervision head
- Model Summary:

```
Total params: 1,944,507
Trainable params: 1,944,507
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 4.00
Forward/backward pass size (MB): 815.66
Params size (MB): 7.42
Estimated Total Size (MB): 827.07
```

Loss Function Analysis

We conducted extensive experiments with multiple loss functions to optimize model convergence and segmentation performance

1. **Dice Loss**

- Advantages:
    - Achieved best final metrics (Dice: 0.90 on validation)
    - Naturally handles class imbalance in cloud coverage

- Challenges:
    - Exhibited training instability (loss oscillations ±15%)
    - Required careful learning rate scheduling

2.**Combination of Dice Loss and Focal Loss**

- Formulation:
    - loss = α*dice_loss + (1-α)*focal_loss , where α=0.7
- Behavior:
    - Improved training stability
    - Smoother gradient flow during early epochs
- Performance Tradeoff:
    - 5-7% lower Dice scores versus pure Dice loss

Thresholding

To get if the predicted pixel in the mask = 1 (contains cloud) we applied a certain threshold other than the typical 0.5 threshold

We made a function to run analysis on the validation dataset to determine the optimal decision boundary for cloud pixel classification

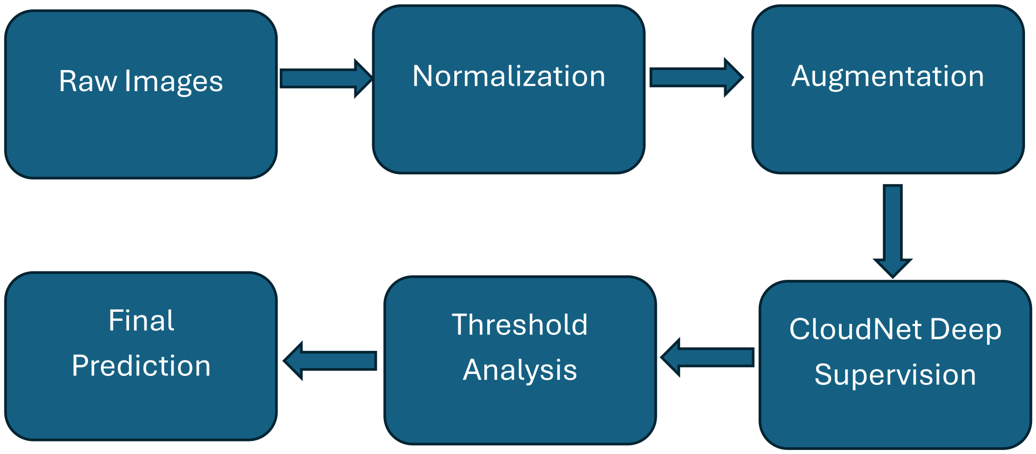
Selection Criteria and Results

We evaluated all variants using dice score on test set

Model Variant	**Test Dice**
CloudNet (Baseline)	0.85
CloudNet Improved	0.87
CloudNet Deep Supervision	0.90

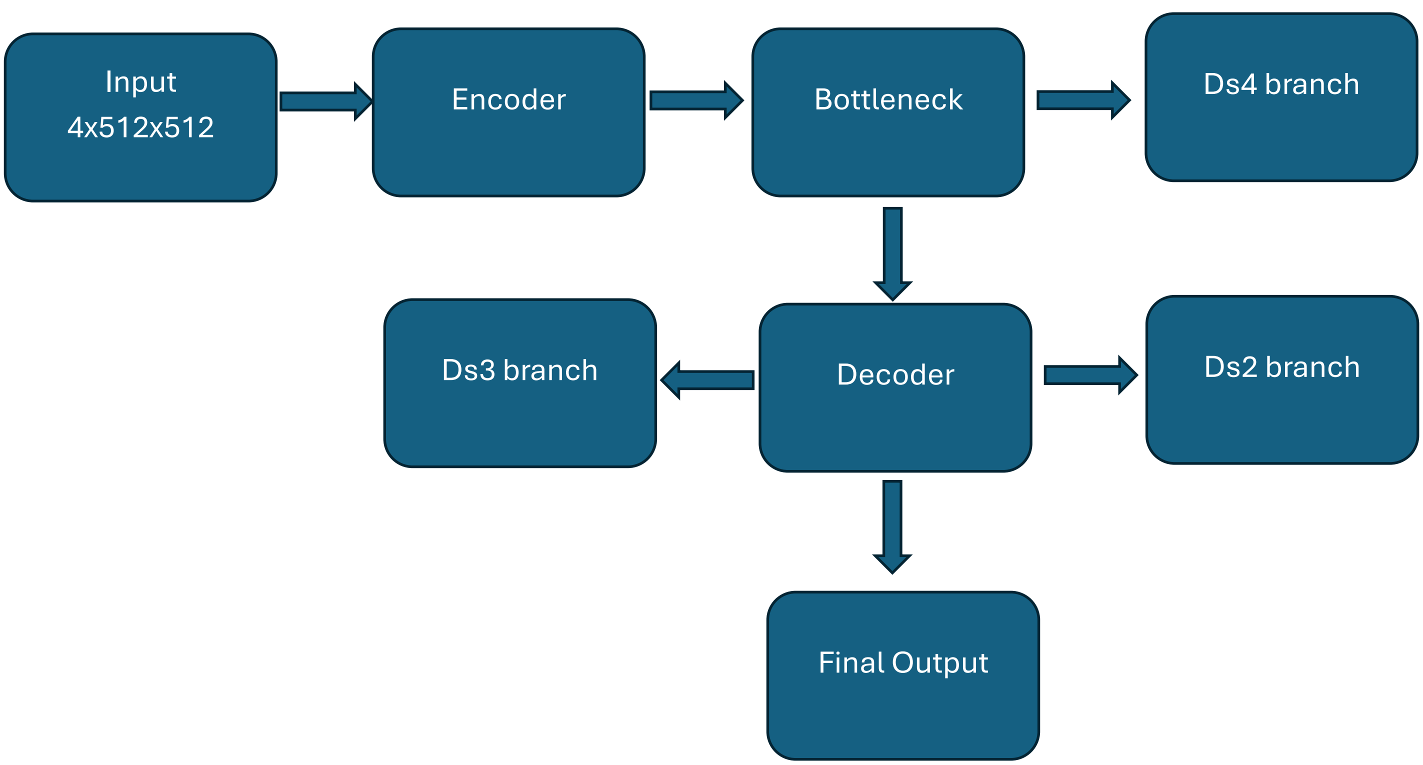
Final Pipeline

We split the data only to train (7779 images) and validation (1945 images)



Training Protocol

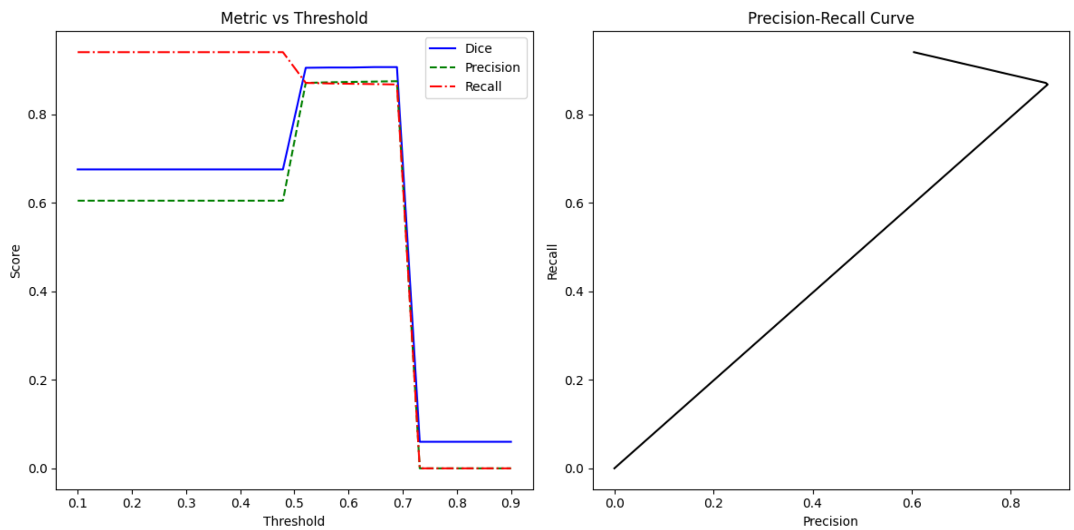
- Model Pipeline



- Optimization:
    - Adam optimizer (lr=1e-3)
    - ReduceLROnPlateau scheduler (factor=0.5, patience=2 epochs)
- Loss Function:

```
# Weighted sum of losses
loss = (1.0 * loss_main +
             0.6 * loss_ds2 + 0.3 * loss_ds3 + 0.1 * loss_ds4)
```

- Number of Parameters: 1,944,507
- Number of Operations (MACs): 2.3367e+10
- Threshold Analysis:



    - The curve shows that the max dice is obtained at threshold = 0.647

Workload Distribution

Names	Responsibility
Ahmed Samy	EDA & Preprocessing
Kareem Samy	Classical models
Nancy Ayman	Deep models
Yara Hisham	Deep models